

Yashasvi Musku

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Education

Master of Science, Computer Science	University of Oklahoma, Norman, OK	Aug 2023 – May 2025
Data Structures, Object-Oriented Design, Algorithm Analysis, Machine Learning, Data Analytics		
Bachelor of Technology, Computer Science	JNTU, Hyderabad, India	Jul 2018 – Jul 2022
Database Management Systems (DBMS), Operating Systems, Computer Networks, SDLC		

Experience

Graduate Research Assistant	University of Oklahoma, Norman, OK	Aug 2024 – May 2025
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- Engineered a **RAG (Retrieval-Augmented Generation) System** using the **LangChain** framework and **OpenAI GPT-4 LLM** to automate faculty content analysis, reducing manual review time by **60%**.
- Implemented an **OCR and embedding pipeline** to process and index large scale documents.
- Optimized **vector-based query matching** in **Pinecone**, reducing average response latency to under **2 seconds**.
- Developed a **full-stack web application** with **Flask** backend and **Tailwind CSS** frontend, integrated **PostgreSQL** for user session management and query logging, achieving **40%** improvement in response accuracy through prompt engineering.

Software Engineer Intern	ISA, University of Oklahoma, Norman, OK	Dec 2023 – Jul 2024
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- Built and optimized a **responsive event and payment portal** using **React**, **Next.js**, **TypeScript**, and **Tailwind CSS** with **REST API** integrations, implementing dynamic forms, client-side validation, and role-based UI components, improving user experience and increasing student event registrations by **35%**.
- Developed and deployed an **AI-powered fraud detection pipeline** leveraging **Isolation Forest**, **scikit-learn**, **Python**, and **FastAPI**, engineering feature extraction, behavioral pattern analysis, and anomaly scoring workflows.

Software Engineer	Aadyam Solutions, Hyderabad, India	Jun 2022 – Jul 2023
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- Contributed to automating the **Aftercare process** for home loan servicing, improving the accuracy of **10,000+ collateral records**, and reducing manual intervention by **40%**.
- Developed and maintained **Spring Boot microservices** to validate mortgage and security data across **3+ verification stages**, enhancing data consistency and reducing processing errors.
- Collaborated on frontend enhancements using **React**, enabling real-time loan status tracking.
- Integrated **PostgreSQL** with **Hibernate/JPA**, ensuring high-integrity financial data retrieval and improving query performance by **30%** through optimized schema design and indexing.
- Executed comprehensive **API testing** via **Postman** and implemented **JUnit-based test coverage**, reducing post-deployment issues by **25%** and ensuring pipeline stability across environments.

Projects

Patient Assistant Network Database System | SQL | T-SQL | Java/JDBC

- Designed ER Diagrams and implemented optimized relational schemas in **Azure SQL Database**, reducing query latency by **30%** via indexing/normalization.
- Built a **Java/JDBC** application to manage CRUD operations, patient data import/export, and team/client financial tracking.

Spring Boot Course Management Application | Spring Boot | AWS | Hibernate | React

- Implemented a full-stack Course Management Application using **Spring Boot**, **Hibernate**, **MySQL**, DAO pattern, and **React** to facilitate student course registration and assignment management.
- Deployed the application on **AWS EC2** with **RDS** for database hosting, enabling scalability and high availability. Provisioned cloud infrastructure using **Terraform**, improving reproducibility and reducing manual configuration.
- Integrated **Apache Kafka** with **AWS Lambda** to asynchronously send email notifications on course enrollments.

Technical Skills

- Programming Languages:** Java, Python, C#, C/C++, JavaScript
- Frameworks/Libraries:** PySpark, Spring Boot, Hibernate, JPA, JUnit, React, Spring MVC, LangChain
- Web/API Technologies:** REST, JSON, OAuth, HTML, CSS, SOAP
- Databases:** PL/SQL, MySQL, PostgreSQL, MongoDB
- Tools/Operating Systems:** Git, Maven, Jenkins, Postman, Jira, Linux
- Cloud/IaC Technologies:** AWS (EC2, Lambda, S3), Google Cloud
- AI/ML & Data Tools:** Apache Spark, Databricks, Pandas, RAG, OCR, Tableau